Lecture 12: Relations Properties.

1- Reflexive.
2- Symmetric
3- Anti Symmetric.

$\forall a,b \in A$ if $(a,b) \in R \wedge (b,a) \in R \rightarrow a = b$

$A = \Phi$ $|A \times A| = |A| \times |A| = 0 \times 0 = 0.$

pow $(A \times A) = \{ \Phi \}.$

$A = \{a\}$ $A \times A = \{(a,a)\}.$

pow $(A \times A) = \{ \Phi, \{(a,a)\}\}.$

$\forall a,b \in A$ if $(a,b) \in R \wedge (b,a) \in R \rightarrow a = b$

$A = \{a,b\}.$ $A \times A = \{(a,a),(a,b),(b,a),(b,b)\}.$

pow $(A \times A) = \{ \Phi, \{(a,a)\}, \{(a,b)\}, \{(b,a)\}, \{(b,b)\},$

$\{(a,a),(a,b)\}, \{(a,a),(b,a)\}, \{(a,a),(b,b)\}, \{(a,b),(b,a)\},$

$\{(a,b),(b,b)\}, \{(b,a),(b,b)\},$

$\{(a,a),(a,b),(b,a)\}, \{(a,a),(a,b),(b,b)\}, \{(a,a),(b,a),(b,b)\},$

$\{(a,b),(b,a),(b,b)\}, \{(a,a),(a,b),(b,a),(b,b)\}\}.$

Transitive: $\forall a,b,c \in A$ if $(a,b) \in R \wedge (b,c) \rightarrow (a,c) \in R.$

$A = \Phi$ $|A \times A| = |A| \times |A| = 0 \times 0 = 0.$

pow $(A \times A) = \{ \Phi \}$

$A = \{a\}$ $A \times A = \{(a,a)\}.$

pow $(A \times A) = \{ \Phi, \{(a,a)\}\}$

$\forall a,b \in A$ if $(a,b) \in R \wedge (b,c) \in R \to (a,c) \in R$.

$A = \{a,b\}$. $\quad A \times A = \{(a,a),(a,b),(b,a),(b,b)\}$.

$Pow(A\times A) = \{\phi, \{(a,a)\}, \{(a,b)\}, \{(b,a)\}, \{(b,b)\},$

$\{(a,a),(a,b)\}, \{(a,a),(b,a)\}, \{(a,a),(b,b)\}, \{(a,b),(b,a)\},$

$\{(a,b),(b,b)\}, \{(b,a),(b,b)\},$

$\{(a,a),(a,b),(b,a)\}, \{(a,a),(a,b),(b,b)\}, \{(a,a),(b,a),(b,b)\},$

$\{(a,b),(b,a),(b,b)\}, \{(a,a),(a,b),(b,a),(b,b)\}$.

$R = \{\phi\}$ ✓✓ $\qquad\qquad A = \{1,2,3,4\}$.

$R = \{(1,1)\}$ ✓✓

$R = \{(1,2),(2,1)\}$ X X.

$\geq\ = HW$.

Ex15 464. $\qquad R = \{(a,b) \mid a \div b \in Z\}$ $\quad (>, <, \leq) \qquad A = Z$.

Reflexive: $\forall a \in A \quad (a,a) \in R$.

$\forall a \in Z. \quad a \div a \in Z$. ✓

Symmetric $\quad \forall a,b \in A$ if $(a,b) \in R \to (b,a) \in R$.

$\forall a,b \in Z$ if $a \div b \in Z \to b \div a \in Z$. X.

$\underbrace{4 \div 2 \in Z}_{T} \to \underbrace{2 \div 4 \in Z}_{F}$.

Anti Symmetric: $\forall a,b \in A$ if $(a,b) \in R \wedge (b,a) \in R \to a=b$.

$\forall a,b \in Z$ if $a \div b \in Z \wedge b \div a \in Z \to a=b$.

Anti symmetric: $\forall a,b \in A$ if $(a,b) \in R \wedge (b,a) \in R \rightarrow a=b$.

$\forall a,b \in \mathbb{Z}$ if $a \div b \in \mathbb{Z} \wedge b \div a \in \mathbb{Z} \rightarrow a = b$. ✓.

Transitive: $\forall a,b,c \in A$ if $(a,b) \in R \wedge (b,c) \in R \rightarrow (a,c) \in R$.

$\forall a,b,c \in \mathbb{Z}$ if $a \div b \in \mathbb{Z} \wedge b \div c \in \mathbb{Z} \rightarrow a \div c \in \mathbb{Z}$. ✓.

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$R_1 = \{(1,1),(2,2),(3,3)\}$.

$R_2 = \{(1,1),(1,2),(1,3),(1,4)\}$.

$R_1 \cap R_2 = \{(1,1)\}$.

$R_1 \cup R_2 = \{(1,1),(2,2),(3,3),(1,2),(1,3),(1,4)\}$.

$R_1 - R_2 = \{(2,2),(3,3)\}$.

$R_2 - R_1 = \{(1,2),(1,3),(1,4)\}$.

$R_1 = \{(x,y) \mid x > y\}$. $A = \mathbb{Z}$.

$R_2 = \{(x,y) \mid x < y\}$.

$R_1 \cap R_2 = \{(x,y) \mid x > y \wedge x < y\} = \emptyset$.

$R_1 \cup R_2 = \{(x,y) \mid x > y \vee x < y\}$.

$= \{(x,y) \mid x \neq y\}$.

$R^{-1} = \{(b,a) \mid (a,b) \in R\}$.

$R = \{(1,2),(2,3),(4,1)\}$.

$R^{-1} = \{(2,1),(3,2),(1,4)\}$.

$\bar{R} = \{(a,b) \mid (a,b) \notin R\}$,

$\bar{R} = A \times A - R$.

$= \{(1,1),(1,3),(2,1),(2,2),(2,3),(3,3)\}$.

$R = \{(1,2),(3,1),(3,2)\}$.

$A = \{1,2,3\}$.

$A \times A = \{(1,1),\cancel{(1,2)},(1,3),(2,1),(2,2),(2,3),\cancel{(3,1)},\cancel{(3,2)},(3,3)\}$,

$R = \{(a,b) \mid a+b=0\}$.

$R^{-1} = \{(b,a) \mid a+b=0\}$,

$\bar{R} = \{(a,b) \mid a+b \neq 0\}$

$R = \{(a,b) \mid a > b\}$ $\quad A = \mathbb{Z}$.

$R^{-1} = \{(b,a) \mid a > b\}$. ✓ $\quad R^{-1} = \{(b,a) \mid (a,b) \in R\}$.

Ex 466-468 (1-30). HW.